

Programming in Java

Activity – Classroom Task on Strings

1. Difference Between String , String Buffer , String Builder

Feature	String	StringBuffer	StringBuilder
Mutability	Immutable (cannot be changed)	Mutable (can be changed)	Mutable (can be changed)
Memory	New object is created whenever modified	Same object is modified	Same object is modified
Performance	Slow for frequent modifications	Faster than String	Fastest among the three
Thread Safety	Safe because immutable	Thread-safe (synchronized)	Not thread-safe (not synchronized)
Synchronization	Not required	Synchronized	Not synchronized
Speed	Slow	Medium	Fast
Used For	Fixed text or constant values	Multithreaded applications	Single-threaded applications

Object Creation	Creates a new object for every modification	No new object for modification	No new object for modification
Memory Usage	More memory due to new objects	Less memory	Less memory
Introduced In	Java 1.0	Java 1.0	Java 1.5
Package	java.lang	java.lang	java.lang
Can Modify Existing Object?	✗ No	✓ Yes	✓ Yes
Best Choice	When content never changes	When multiple threads modify text	When a single thread modifies text frequently

2.String Class Methods and Description

Method	Description
length()	Returns the number of characters in the string.
charAt(int index)	Returns the character at the specified index.
substring(int beginIndex)	Returns a substring from the given index to the end.
substring(int beginIndex, int endIndex)	Returns a substring between the specified indices.
contains(CharSequence s)	Checks whether the string contains the specified sequence.
equals(Object obj)	Compares two strings for equality.
equalsIgnoreCase(String str)	Compares two strings ignoring case differences.
compareTo(String str)	Compares two strings lexicographically.
compareToIgnoreCase(String str)	Compares two strings ignoring case.
concat(String str)	Concatenates the specified string to the end.
isEmpty()	Returns true if the string is empty.

isBlank()	Returns true if the string is empty or contains only whitespace (Java 11+).
startsWith(String prefix)	Checks if the string starts with the specified prefix.
endsWith(String suffix)	Checks if the string ends with the specified suffix.
indexOf(String str)	Returns the first occurrence index of the specified string.
indexOf(String str, int fromIndex)	Returns the first occurrence from the specified index.
lastIndexOf(String str)	Returns the last occurrence index of the specified string.
lastIndexOf(String str, int fromIndex)	Returns the last occurrence before the specified index.
replace(char oldChar, char newChar)	Replaces all occurrences of a character.
replace(CharSequence target, CharSequence replacement)	Replaces all matching character sequences.
replaceAll(String regex, String replacement)	Replaces all substrings matching a regular expression.

replaceFirst(String regex, String replacement)	Replaces the first substring matching a regular expression.
split(String regex)	Splits the string into an array using a delimiter.
split(String regex, int limit)	Splits the string with a specified limit.
toUpperCase()	Converts the string to uppercase.
toLowerCase()	Converts the string to lowercase.
trim()	Removes leading and trailing whitespace.
strip()	Removes leading and trailing Unicode whitespace (Java 11+).
stripLeading()	Removes leading whitespace.
stripTrailing()	Removes trailing whitespace.
toCharArray()	Converts the string into a character array.
getBytes()	Converts the string into a byte array.
getChars(int srcBegin, int srcEnd, char[] dst, int dstBegin)	Copies characters into a character array.
matches(String regex)	Checks whether the string matches a regular expression.

intern()	Returns the canonical representation from the string pool.
repeat(int count)	Repeats the string the specified number of times (Java 11+).
lines()	Returns a stream of lines from the string (Java 11+).
subSequence(int start, int end)	Returns a character sequence between indices.
codePointAt(int index)	Returns the Unicode code point at the specified index.
codePointBefore(int index)	Returns the Unicode code point before the specified index.
codePointCount(int begin, int end)	Counts Unicode code points in the specified range.
offsetByCodePoints(int index, int offset)	Returns the index offset by Unicode code points.
valueOf() (static)	Converts different data types into a String.
copyValueOf(char[] data) (static)	Creates a string from a character array.
format(String format, Object... args) (static)	Returns a formatted string.

join(CharSequence delimiter, CharSequence... elements) <i>(static)</i>	Joins strings using a delimiter.
toString()	Returns the string itself.

3. String Buffer Methods and Description

Method	Description
append()	Appends data to the end of the string.
insert(int, data)	Inserts data at the specified index.
delete(int, int)	Deletes characters between the specified indices.
deleteCharAt(int)	Deletes the character at the specified index.
replace(int, int, String)	Replaces characters between the specified indices.
reverse()	Reverses the characters in the string.
length()	Returns the number of characters.
capacity()	Returns the current capacity of the buffer.
ensureCapacity(int)	Ensures the minimum capacity.
trimToSize()	Reduces capacity to match the current length.
setLength(int)	Sets the length of the buffer.
charAt(int)	Returns the character at the specified index.
setCharAt(int, char)	Replaces the character at the specified index.

substring(int)	Returns a substring from the specified index.
substring(int, int)	Returns a substring between two indices.
subSequence(int, int)	Returns a character sequence.
indexOf(String)	Returns the first occurrence index of a string.
indexOf(String, int)	Returns the first occurrence from a specified index.
lastIndexOf(String)	Returns the last occurrence index of a string.
lastIndexOf(String, int)	Returns the last occurrence before a specified index.
getChars(int, int, char[], int)	Copies characters into a character array.
appendCodePoint(int)	Appends a Unicode code point.
codePointAt(int)	Returns the Unicode code point at the specified index.
codePointBefore(int)	Returns the Unicode code point before the specified index.
codePointCount(int, int)	Counts Unicode code points in a range.
offsetByCodePoints(int, int)	Returns the index offset by Unicode code points.
toString()	Converts the buffer to a String.
isEmpty()	Returns true if the buffer is empty (Java 15+).
compareTo(StringBuffer)	Compares two StringBuffer objects lexicographically (Java 11+).

4. String Builder Methods and Description

Method	Description
append()	Appends data to the end of the string.
insert(int, data)	Inserts data at the specified index.
delete(int, int)	Deletes characters between the specified indices.
deleteCharAt(int)	Deletes the character at the specified index.
replace(int, int, String)	Replaces characters between the specified indices.
reverse()	Reverses the characters in the string.
length()	Returns the number of characters.
capacity()	Returns the current capacity.
ensureCapacity(int)	Ensures the minimum capacity.
trimToSize()	Reduces capacity to the current length.
setLength(int)	Sets the length of the builder.
charAt(int)	Returns the character at the specified index.
setCharAt(int, char)	Replaces the character at the specified index.
substring(int)	Returns a substring from the specified index.
substring(int, int)	Returns a substring between two indices.
subSequence(int, int)	Returns a character sequence.
indexOf(String)	Returns the first occurrence index of a string.

indexOf(String, int)	Returns the first occurrence from a specified index.
lastIndexOf(String)	Returns the last occurrence index of a string.
lastIndexOf(String, int)	Returns the last occurrence before a specified index.
getChars(int, int, char[], int)	Copies characters into a character array.
appendCodePoint(int)	Appends a Unicode code point.
codePointAt(int)	Returns the Unicode code point at the specified index.
codePointBefore(int)	Returns the Unicode code point before the specified index.
codePointCount(int, int)	Counts Unicode code points in a range.
offsetByCodePoints(int, int)	Returns the index offset by Unicode code points.
toString()	Converts the builder to a String.
isEmpty()	Returns true if the builder is empty (Java 15+).
compareTo(StringBuilder)	Compares two StringBuilder objects lexicographically (Java 11+).